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1. Define cloud computing and explain any four characteristics.

* on-demand Self-Service

Users can access resources wherever required.

* Broad network access:

Services are available through the internet using computers and mobile devices.

* Resource pooling

Computing resources are shared among multiple users.

* Rapid elasticity.

Resources can be increased or decreased according to demand.

Trace the evolution of cloud computing.

1. mainframe computing
Large computers served multiple users.

2. Personal computers
computing became available to individuals.

3. Internet development
users could access information and services online.

4. Virtualization.
multiple virtual machines could run on one physical server.

5. web services.
Applications and resources became accessible through the internet.

6. modern cloud platforms.
Providers such as AWS, microsoft Azure, and Google cloud offer scalable computing storage, and software services.

Explain the role of server virtualization in cloud computing. Differentiate virtualization and cloud computing.

Server virtualization.

It allows a single physical server to run multiple virtual machines using a hypervisor. Each virtual machine can run its own operating system and applications.

Difference:

* virtualization: A technology that divides physical resources into virtual resources.

* cloud computing: A service model that delivers computing resources over a network on demand.

Discuss the role of data centers in delivering cloud services.

Data centers are facilities containing servers, storage systems, networking equipment, and security systems. They play an important role in cloud computing by:

1. Storing and processing large amounts of data
2. Providing computing resources and applicat
3. Ensuring data availability through backup and recovery.
4. maintaining security and controlling access.
5. Providing reliable services using power backup, cooling, and network connectivity.

...SaaS, and XaaS

are example each.

model	meaning	Description
IaaS	Infrastructure as a Service	Provides virtual servers, storage, and networking.
PaaS	Platform as a Service	Provides a platform to develop and deploy applications.
SaaS	Software as a Service	Provides ready-use software over internet
XaaS	Anything as a Service	Delivers various IT resources as services.

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